

MODULE 05

Exploratory Data Analysis EDA

AI with Data Science - Professional Learning Series

ARSHITH LMS | Professional Edition

Exploratory Data Analysis EDA

What is EDA?

EDA is the process of analysing data to summarise its main characteristics using statistics and visualisations before modelling.

Goals:

- Understand data distributions
- Find patterns and relationships
- Detect outliers and anomalies
- Test assumptions
- Generate hypotheses

Univariate Analysis

```
import pandas as pd, matplotlib.pyplot as plt, seaborn as sns

# Numerical features
df['age'].describe()
plt.hist(df['age'], bins=20, edgecolor='black')
sns.boxplot(x=df['age'])

# Categorical features
df['city'].value_counts()
df['city'].value_counts().plot(kind='bar')
sns.countplot(x='city', data=df)
```

Bivariate Analysis

```
# Numerical vs Numerical
plt.scatter(df['age'], df['salary'])
sns.regplot(x='age', y='salary', data=df)

# Categorical vs Numerical
sns.boxplot(x='department', y='salary', data=df)
sns.violinplot(x='department', y='salary', data=df)

# Categorical vs Categorical
pd.crosstab(df['gender'], df['department'])
sns.heatmap(pd.crosstab(df['gender'], df['dept']), annot=True)
```

Multivariate Analysis

```
import seaborn as sns

# Pair plot
sns.pairplot(df, hue='category')
```

```

# Correlation heatmap
import matplotlib.pyplot as plt
corr = df.corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', center=0)
plt.title('Feature Correlation Matrix')
plt.show()

# 3D scatter
from mpl_toolkits.mplot3d import Axes3D
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
ax.scatter(df['x'], df['y'], df['z'])

```

EDA Insights and Reporting

After EDA, document findings:

1. Data Quality: missing values, duplicates, wrong types
2. Distributions: skewness, outliers, normality
3. Relationships: strong correlations (potential multicollinearity)
4. Class Imbalance: unequal target class distribution
5. Feature Importance: which features seem most predictive

Use pandas-profiling for automated EDA:

```

import pandas_profiling
profile = df.profile_report()
profile.to_file('eda_report.html')

```